

(a) Restriction endonucleases are used in genetic modification. These enzymes occur naturally in prokaryotic cells. More than 3500 different restriction endonucleases have been identified and it is thought there are many more to discover.

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- This original method produced only a small quantity of restriction endonuclease and was **not** economical.

- obtain the gene coding for a specific restriction endonuclease
- introduce the gene into *Escherichia coli*, with a promoter that allows the gene to be expressed continuously.

Suggest **and** explain the steps needed to carry out the newer method for large-scale production of a specific restriction endonuclease.

[4]

(c) Describe the advantages of databases for the study and use of restriction endonucleases.

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(d) Electrophoresis is a technique used in genetic technology.

Paper chromatography is a technique used to investigate the photosynthetic pigments found in chloroplasts.

Compare the similarities and differences between electrophoresis and chromatography.

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